

LANGARA, BC, Canada

WMO: 718990

Lat: 54.255N

Lon: 133.058W

Elev: 140

StdP: 14.62

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	25.3	28.3	12.8	10.6	28.0	17.3	13.3	30.0	37.6	38.3	34.9	38.4	15.0	90	0.710

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.8	63.1	58.9	61.3	58.0	59.9	57.2	60.0	62.0	58.7	60.3	57.7	59.3	8.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.0	75.0	60.5	57.9	72.0	59.4	56.9	69.4	58.5	26.4	62.2	25.6	60.5	25.0	59.4	65.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.9	27.2	23.3	21.9	69.2	4.9	5.5	18.3	73.1	15.5	76.3	12.7	79.4	9.1	83.4
			19.7	62.1	4.6	1.8	16.3	63.4	13.6	64.4	11.0	65.4	7.7	66.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	39.5	39.4	39.7	43.4	47.4	51.6	55.1	57.0	54.6	48.5	42.5	39.5
	DBStd	7.39	4.98	3.99	4.29	3.07	2.96	2.79	2.22	2.17	2.59	3.32	4.57	4.69
	HDD50	1861	325	296	319	201	92	11	0	0	1	66	225	325
	HDD65	6734	790	716	784	649	547	402	307	249	313	512	675	790
	CDD50	605	0	0	0	2	10	60	158	217	138	19	1	0
	CDD65	1	0	0	0	0	0	1	0	0	0	0	0	0
	CDH74	7	0	0	0	0	0	6	0	1	0	0	0	0
	CDH80	1	0	0	0	0	0	1	0	0	0	0	0	0
	WSAvg	10.9	12.8	11.9	12.0	10.2	9.7	8.7	8.8	8.7	10.1	12.0	12.9	13.2
	PrecAvg	76.9	7.8	5.9	5.8	5.4	4.2	3.8	3.5	4.7	7.3	10.9	8.9	8.7
(15) Precipitation	PrecMax	94.3	14.1	10.2	10.8	11.6	10.5	6.1	8.0	10.1	12.3	16.8	15.8	15.9
	PrecMin	58.1	2.6	0.7	0.9	2.1	1.3	1.5	0.9	1.0	2.5	5.2	3.5	3.0
	PrecStd	8.8	2.5	2.2	2.1	2.0	1.9	1.4	1.7	2.3	2.5	2.7	2.8	2.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.0	49.8	50.4	56.8	60.6	65.8	63.6	66.5	64.0	57.8	53.0	50.2
		MCWB	48.2	47.3	47.3	48.3	50.9	56.4	59.3	60.6	60.0	55.7	51.1	49.0
	2%	DB	48.1	47.5	48.2	51.8	55.6	59.2	61.2	63.6	61.3	55.7	51.1	48.3
		MCWB	47.0	45.6	45.8	47.3	51.1	55.2	58.0	59.7	58.2	53.4	49.7	47.2
	5%	DB	46.8	45.9	46.6	49.4	53.3	57.2	59.7	61.9	59.6	54.2	49.5	46.9
		MCWB	45.6	44.2	44.5	46.1	49.8	53.9	57.1	58.7	57.1	52.2	48.2	46.1
	10%	DB	45.5	44.6	45.1	47.8	51.6	55.4	58.4	60.5	58.4	53.0	48.3	45.7
		MCWB	44.4	43.1	43.3	45.0	48.7	52.7	56.3	57.9	56.2	51.1	47.0	44.8
	0.4%	WB	49.0	48.3	48.5	50.6	53.9	57.6	60.2	62.1	61.5	56.8	52.0	49.4
		MCDB	49.3	49.1	49.5	53.8	56.1	62.3	62.2	64.5	62.8	57.5	52.4	49.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.3	46.1	46.6	48.3	51.8	55.6	58.5	60.5	59.2	54.3	50.2	47.6
		MCDB	47.8	47.0	47.7	50.7	54.4	58.2	60.2	62.7	60.4	55.1	50.8	48.1
	5%	WB	45.9	44.6	45.0	46.8	50.3	54.3	57.6	59.2	57.9	52.9	48.6	46.3
		MCDB	46.6	45.6	46.1	48.9	52.7	56.5	59.1	61.0	59.0	53.7	49.3	46.7
	10%	WB	44.6	43.3	43.7	45.5	49.2	53.1	56.6	58.2	56.7	51.5	47.3	44.8
		MCDB	45.3	44.3	44.9	47.6	51.3	55.0	58.0	59.8	57.9	52.5	48.1	45.5
	5% DB	MDBR	5.2	5.7	7.1	8.0	7.4	7.1	6.1	6.8	6.9	5.9	5.5	5.2
		MCDBR	6.0	6.5	8.2	9.7	9.8	9.7	7.8	9.0	8.3	7.1	6.1	5.7
		MCWBR	5.8	5.7	6.5	7.0	6.8	6.4	5.6	6.2	6.1	5.9	5.8	5.8
		MCWBR	5.9	5.9	7.5	9.0	8.8	8.5	7.3	8.0	7.5	6.6	6.1	5.9
(35) Mean Daily Temperature Range	5% WB	MCWBR	5.9	5.6	6.7	6.9	6.6	6.2	5.6	5.9	6.2	6.2	6.1	6.1
(40) Clear-Sky Solar Irradiance	taub		0.312	0.319	0.343	0.382	0.378	0.371	0.374	0.372	0.370	0.347	0.317	0.305
	taud		2.367	2.481	2.426	2.303	2.386	2.448	2.486	2.506	2.452	2.417	2.336	2.294
	Ebn at Noon		204	249	266	268	276	279	275	270	253	231	197	177
	Edh at Noon		17	22	28	36	35	33	32	30	28	23	18	15
(44) All-Sky Solar Radiation	RadAvg		161	413	756	1210	1477	1568	1404	1221	853	468	221	128
	RadStd		19	67	90	59	107	121	128	110	85	51	24	19

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page